S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering - RWTH

+49 17685944047 25/04/2002 Renningen, Stuttgart

vallabh.pataneni@rwth-aachen.de Vallabh-Pataneni

Summary

A highly motivated Automotive Engineering Master's student focused on optimizing complex vehicle systems by merging **Systems** and **Industrial Engineering** principles with **Artificial Intelligence**. Architecting and documenting complex, rule-based operating strategies at Robert Bosch GmbH, grounded in rigorous **scientific writing** and analytical methodologies. Combines leadership in **end-to-end vehicle development** with hands-on experience in **process analysis** and data-driven optimization in Python. Seeking a **Thesis** position to apply these skills to optimize **requirements management** and shape the future of sports car development at Porsche.

Experience

System Simulation Intern, *Robert Bosch GMBH* 09/2025 – Present | Schwieberdingen

- Modeled and calibrated a PoC rule-based powertrain strategy in **Simulink**, and developed a new strategy in a **MATLAB script** for minimum fuel consumption and mitigation of fuel cell degradation. Performed multi-objective optimizations using **machine learning** with **Genetic Algorithms** and **Gradient Descent** to generate control MAPs, benchmarking against industry standards (DP, ECMS). Formalized this state-of-the-art strategy in a peer-reviewed manuscript.
- **Key Skills:** MATLAB/Simulink, Machine learning, Gradient Descent, Multi-Objective Optimization, PoC Validation, Systems Engineering, Scientific Writing, Process Analysis.

Technical Team Lead, *Ecogenium e.V - Student Team* 06/2024 – Present | Aachen

- Led the end-to-end development of a hydrogen fuel-cell vehicle, deploying data-driven strategies in **Python/Simulink** and **Power BI** to analyze competition data and identify performance bottlenecks. Built and validated **Simulink**-based models for motor testing and maximized vehicle efficiency through Multi Body Simulations. Spearheaded the **analysis of existing processes** and methods to improve overall vehicle efficiency and performance.
- **Key Skills:** Python, Simulink, CATIA V5, ANSYS, Kinematic & Structural Analysis,MSC ADAMS,Process Analysis, Project Leadership.

Industrial Engineering Intern, *Techolution PVT LTD* 05/2023 – 08/2023 | Hyderabad

- Led the **end-to-end development of a user-centric mechatronic system**, from design and simulation to prototyping and testing. Engineered a complete **digital twin** to validate control logic and authored a **MATLAB** motor-control script to emulate human-like motion, streamlining the 3rd-gen robot's release with **Fusion 360 Manage**.
- **Key Skills:** Python, Fusion 360, CoppeliaSim, Matlab, Digital Twin, Virtual Commissioning.

Skills

Simulation-Proficiency

- Matlab/Simulink
- MSC ADAMS
- CoppeliaSIM

Software-Proficiency

- PYTHON
- C/C++ (Arduino IDE)
- MS Office Suit

PLM Software - Proficiency

- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Data Analysis - Proficiency

Power Bi

Tableau

Education

Master of Science in Automotive Engineering, 09/2023 – Present | Aachen, Germany
RWTH Aachen University

Relevant Coursework: Automotive Dynamics and Components Simulation Sciences, Mechatronics,Fundamentals of Machine Learning.

GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity

GPA: 1.3

Languages

English
Fluent (C1)

German
Intermediate (B1)

Academic Achievements

Lightweight Foldable Electric Wheelchair, *ISEA - RWTH* 02/2025

Pitched and co-developed the initial concept for a lightweight, foldable electric wheelchair with a user-removable Li-ion battery pack. Engineered the complete battery pack architecture and developed a foundational Simulink model to simulate discharge curves and estimate vehicle range after conducting a comprehensive analysis to select the optimal battery cell.

Academic Recognization, *RWTH* 10/2024

Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.3 in core automotive engineering courses.

Paper Publication, *IJSED Research* 11/2022

A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Declaration

I hereby declare that the information provided in this resume is true and accurate to the best of my knowledge.

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Summary

A highly motivated Automotive Engineering Master's student focused on architecting intelligent vehicle systems by merging dynamic modeling with **machine learning**. Already a contributor to **Robert Bosch GmbH's** innovation, developing a smart powertrain strategy optimized with **Gradient Descent**. Combines this with **leadership** in **end-to-end vehicle development**, hands-on creation of **digital twins**, and experience in the scientific publication process. Seeking a Thesis position to apply these skills in **simulation** and **machine learning** to build innovative **compositional models** for **electric drives** at Bosch

Experience

System Simulation Intern, *Robert Bosch GMBH* 09/2025 – Present | Schwieberdingen

- Modeled and calibrated a PoC rule-based powertrain strategy in **Simulink**, and developed a new strategy in a **MATLAB script** for minimum fuel consumption and mitigation of fuel cell degradation. Performed multi-objective optimizations using **machine learning** with **Genetic Algorithms** and **Gradient Descent** to generate control MAPs, benchmarking against industry standards (DP, ECMS). Formalized this state-of-the-art strategy in a peer-reviewed manuscript.
- **Key Skills:** MATLAB/Simulink,Machine learning, Gradient Descent, Multi-Objective Optimization,PoC Validation, Benchmarking,Literature Review.

Technical Team Lead, *Ecogenium e.V - Student Team* 06/2024 – Present | Aachen

- Led the end-to-end development of a hydrogen fuel-cell vehicle, deploying data-driven strategies in **Python/Simulink** and **Power BI** to analyze competition data and identify performance bottlenecks.that **boosted fuel efficiency by ~80%**. Built and validated **Simulink**-based models for motor testing and their operating curves, and maximized vehicle efficiency through Multi Body Simulations.
- **Key Skills:** Python, Simulink, CATIA V5, ANSYS, Kinematic & Structural Analysis,MSC ADAMS,Leadership.

Industrial Design Intern, *Techolution PVT LTD* 05/2023 – 08/2023 | Hyderabad

- Led the end-to-end development of a user-centric DoorOpen robot, from design and simulation in **Fusion 360** to prototyping and testing. Engineered a complete **digital twin** for a factory automation cell to validate control logic and authored a motor-control script to emulate human-like motion, streamlining the 3rd-gen robot's release with **Fusion 360 Manage**.
- **Key Skills:** Python, Fusion 360, CoppeliaSim, Matlab, Digital Twin, Virtual Commissioning.

Skills

Simulation-Proficiency

- Matlab/Simulink
- MSC ADAMS
- CoppeliaSIM

PLM Software - Proficiency

- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Python-Libraries

- NumPy
- Pandas

Data Analysis - Proficiency

Power Bi
Tableau

Education

Master of Science in Automotive Engineering, 09/2023 – Present | Aachen, Germany
RWTH Aachen University

Relevant Coursework: Automotive Dynamics and Components Simulation Sciences, Mechatronics,Fundamentals of Machine Learning.

GPA: 1.7

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Summary

A motivated Automotive Engineering Master's student with a deep **passion for hydrogen mobility**. Possesses a comprehensive understanding of **Fuel Cell** technology, from a sophisticated **system-level** perspective gained at **Robert Bosch GmbH** creating advanced **fuel cell EV operating strategies** in MATLAB/Simulink focusing on **Fuel Consumption** and **Fuel Cell Degradation**, to a **fundamental and operational** understanding from **Ecogenium e.V**. This was gained by leading **Germany's only student-built fuel cell car** team, managing the **complete vehicle development lifecycle** from concept, design, and simulation to prototyping and fabrication, complemented by **extensive fuel cell testing** to improve performance. Eager to blend this unique expertise in fuel cell technology to contribute to the pioneering fuel cell propulsion system projects at **DLR**

Experience

System Simulation Intern, *Robert Bosch GMBH* 09/2025 – 12/2025 | Schwieberdingen

- **Integrated Powertrain Strategy & Modeling:** Developed and validated an iterative Simulink model for a rule-based powertrain operating strategy in the EVsim module for Fuel Cell PHEVs, Validating its **Fidelity** with measurement data.
- **Battery & Fuel Cell Degradation Mitigation:** Engineered a new MATLAB-based powertrain strategy to safeguard against Fuel Cell and HV-Battery degradation by managing dynamic loads, thermal stress, low voltage operation, PE Ratios, SOC swing, and life cycles.
- **Multi-Objective Optimization:** Executed multi-variable parametric optimization on large datasets to generate lookup tables (MAPs) for optimal recuperation, charge sustenance, and minimal fuel consumption.
- **KPI Analysis & Documentation:** Developed post-processing plots and technical documents to analyze and benchmark efficiency and degradation KPIs (Power Distribution, SOC window, etc.) against DP and ECMS standards.
- **Scientific Publication & Research:** Currently **Co-authoring a scientific paper** detailing the novel application and effectiveness of the **State-of-the-Art Rule-Based Powertrain Control Strategies** developed during the internship.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric DoorOpen robot, focusing on its **dynamics** and **mechanical reliability** using iterative simulation strategies in Fusion 360.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Door-open Robot** by creating a single source of truth in **Autodesk Fusion 360 Manage**, which integrated 3D CAD models, electronic datasheets, user manuals, FEM/thermal simulation results, and manufacturing drawings.
- Engineered a **data-driven** motor control script in **MATLAB**, correlating real-world **sensor measurement data** with a control model to successfully replicate human-centric **acceleration/deceleration profiles**.
- Utilized **rapid prototyping** (Raise 3D) for the fabrication and continuous design refinement of over 150 parts.

Student Club

Ecogenium e.V, *Technical Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- **Vehicle Dynamics & Efficiency Simulation:** Optimized **longitudinal and lateral dynamics** to maximize efficiency by performing kinematic simulations of **Front** and **Rear Suspension** in MSC ADAMS to determine the optimal coordinates, minimizing rolling resistance and improving vehicle performance.
- **Suspension Development**: **Engineered** the complete front **double wishbone** and rear **trailing arm suspension systems** and its components using **Catia V5** and **Siemens NX**,applying **Generative Design**, **topology optimization,** and **FEM simulation** to achieve efficient packaging and minimal weight.
- **Fuel Cell Test Bench Development**:Engineered, fabricated, and electronically integrated custom-built test benches (standalone fuel cell, quarter-model suspension) to conduct performance data evaluation and validate simulation models against real-world hardware data.
- **Performance Characterization & Analysis:** Gathered extensive operational data through a wide range of performance-based experiments, including polarization curves to quantify voltage losses, as well as purging, short-circuiting, and humidity cycling tests and mapped the stack's optimal thermal operating window by adjusting fan speeds.
- **Energy Management & Strategy:** Architected and deployed advanced **Simulink** and Python-based energy management strategies for a hydrogen fuel cell vehicle. This included real-time control of the fuel cell's temperature and power output, boosting on-track fuel efficiency by **80%** and securing a runner-up finish at the Shell Eco-Marathon.

Education

Master of Science in Automotive Engineering, 09/2023 – Present | Aachen, Germany
RWTH Aachen University
Relevant Coursework: Automotive Engineering Dynamics, Components and Assistant Systems, Vehicle Structures, Mechatronics, Simulation Sciences, Finite Element Methods (FEM), CAD/CAM, DFM, DfAM.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Academic Project

Electric Wheelchair with Modular Battery System, 10/2024 – 02/2024
ISEA-RWTH
- **Pitched and co-developed the initial concept** for a lightweight, foldable electric wheelchair, which featured a user-removable Li-ion battery pack.
- Conducted a **comprehensive analysis** to **select the optimal battery cell**, evaluating critical metrics including energy density, use case,thermal stability, weight, and cycle lifetime.
- **Engineered the complete battery pack architecture**, including **power electronics** and charging strategy, and developed a foundational **Simulink model** to **simulate discharge curves** and **estimate vehicle range.**

Skills

CAD-Proficiency
- Catia V5
- NX SIEMENS
- FUSION 360
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Matlab/Simulink
- MSC ADAMS
- ANSYS STRUCTURAL , FLEUNT , ACP
- DYMOLA
- Solidworks

Software-Proficiency
- PYTHON
- C/C++
- MS Office Suit

PLM Software - Proficiency
- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Academic Recognization, *RWTH*
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.3 in core automotive engineering courses.

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Summary

As an Automotive Engineering Master's student, my career is purposefully aimed at advancing **hydrogen mobility**. I have built a unique, holistic expertise that connects industrial-grade simulation with hands-on application. At **Robert Bosch GmbH**, I am developing **FCEV operating strategies** in **MATLAB/Simulink** to improve Fuel efficiency while safeguarding fuel cell degradation. Concurrently, as a leader at **Ecogenium e.V.**, I am managing the ground-up build of Germany's only **Student hydrogen fuel cell vehicle**. This is complemented by proven skills in **techno-economic (TCO) analysis** and the **full component lifecycle**, from CAD,simulation to prototyping and validation. I am seeking this Master Thesis to apply my direct FCEV strategy and systems experience to Fraunhofer IAO's research in **'Sustainable solutions for future mobility and energy transition'** the **'Development and implementation of innovation and technology strategies'**.

Education

Master of Science in Automotive Engineering, 09/2023 – Present | Aachen, Germany

RWTH Aachen University

Relevant Coursework:Alternative Propulsion Systems, Thermodynamics, Mechatronics, Simulation Sciences, Material Science, Energy Transition.

GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India

Jawaharlal Nehru Technological Univerisity

GPA: 1.3

Student Club

Ecogenium e.V, *Mechanical Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- **Suspension Development: Engineered** the complete **front (double wishbone)** and **rear (trailing arm) suspension systems** and its **Components** using **Catia V5** and **Siemens NX**, performing **kinematic analysis in ADAMS** and applying **Generative Design**, **topology optimization**, and **FEM simulation** to achieve efficient packaging and minimal weight.
- **Prototyping & Validation:** Prototyped topology-optimized suspension components using **Metal 3D Printing (DMLS)**, then validated them on a custom-built, **quarter-model test bench** to conduct **movement and load testing** against **ADAMS** simulation data.
- **Fuel Cell Test Bench Development:** Engineered, fabricated, and electronically integrated a standalone **fuel cell test bench**, executing operational testing (purging, humidity cycling, short-circuiting) and performance data evaluation.
- **Strategy Implementation** : Developed and implemented **data-driven driving and control strategies** using **MATLAB** and **Simulink**, leveraging competition data to improve **fuel efficiency by 80%** and help secure a **runner-up finish** at the Shell Eco-Marathon.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 09/2025 – 12/2025 | Schwieberdingen

- Developed and integrated a iterative **Simulink** model for an existing power-split operating strategy concept into the EVsim (PHEV) module, **validating its fidelity** against reference Excel data.
- Engineered a new, elaborate operating strategy in a **MATLAB script** for Strong **Fuel Cell** Hybrid EVs (SEVs), based on the initial concept; this new strategy focuses on **safeguarding Fuel Cell degradation** and is parameterized by various MAPs (lookup tables) for different battery capacities and drive cycles.
- Executed **Multi-objective,multi-variable parametric optimization** to generate the required MAPs, specifically targeting **charge sustenance** and **minimal fuel consumption**.
- Developed **post-processing plots** to analyze **efficiency** and **degradation KPIs** , **benchmarking** the new strategy's performance against DP and ECMS.
- Currently **co-authoring a scientific paper** based on our State of the Art FCEV strategy.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric DoorOpen robot, focusing on its **dynamics** and **mechanical reliability** using iterative simulation strategies in Fusion 360.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Door-open Robot** by creating a single source of truth in **Autodesk Fusion 360 Manage**, which integrated 3D CAD models, electronic datasheets, user manuals, FEM/thermal simulation results, and manufacturing drawings.
- Developed and built a custom **test rig** to evaluate the **clamping performance** of a **robot's Chassis** across varied surfaces (steel, aluminum, carbon fiber) to optimize **material selection and reduce weight** .
- Utilized **rapid prototyping (Raise 3D)** for the fabrication and continuous design refinement of over 150 parts.

Academic Project

Electric Wheelchair with Modular Battery System, 10/2024 – 02/2024
ISEA-RWTH

- **Pitched and co-developed the initial concept** for a lightweight, foldable electric wheelchair, which featured a user-removable Li-ion battery pack.
- Conducted a **comprehensive analysis** to **select the optimal battery cell**, evaluating critical metrics including energy density, use case,thermal stability, weight, and cycle lifetime.
- **Engineered the complete battery pack architecture**, including **power electronics** and charging strategy, and developed a foundational **Simulink model** to **simulate discharge curves** and **estimate vehicle range.**

Well-to-Wheel (WTW) & TCO Analysis of EVs vs. ICEs, 01/2022 – 06/2022
JNTUHCEH

- Conducted an **extensive literature review** on existing WTW analyses, then **modified key parameters** (e.g., energy mix, consumer costs) to create a model for the **Indian market**.
- Performed a comprehensive **Well-to-Wheel (WTW) sustainability analysis** based on this model to critically evaluate regional "zero-emission" claims by assessing local energy generation.
- Executed a **Total Cost of Ownership (TCO)** and **amortization analysis** to compare the lifetime affordability of EVs and ICEs, providing a complete, region-specific economic and environmental picture.

Skills

CAD-Proficiency
- Catia V5
- NX SIEMENS
- FUSION 360
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Matlab/Simulink
- ANSYS STRUCTURAL & FLEUNT
- DYMOLA
- Solidworks
- Siemens NX

Software-Proficiency
- PYTHON
- C/C++
- MS Office Suit

PLM Software - Proficiency
- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Academic Recognization, *RWTH*
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.3 in core automotive engineering courses.

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Summary

As an Automotive Engineering Master’s student, my career is purposefully aimed at advancing **hydrogen fuel cell mobility** toward commercial viability. This focus is demonstrated through two distinct perspectives: 'top-down' industrial simulation at **Bosch**, where I developed strategies to **safeguard fuel cell degradation**, and 'bottom-up' practical leadership at **Ecogenium e.V.**, where I guided the ground-up development of a complete FCEV. I possess proven, end-to-end experience across the **complete life cycle of vehicle components**: from conceptualization and digital analysis through load and kinematic analysis to prototyping with **advanced manufacturing** techniques and validation on **custom test benches**. I am now seeking this thesis to transition from system-level optimization to the foundational, hands-on work of developing, formulating, and optimizing the core components I believe in.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework:Material Science, Production Engineering, Additive Manufacturing, Alternative Propulsion Systems, Thermodynamics, Design for Manufacturing (DFM), FEM.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

Ecogenium e.V, *Mechanical Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- **Suspension Development: Engineered** the complete **front (double wishbone)** and **rear (trailing arm) suspension systems** and its **Components** using **Catia V5** and **Siemens NX**, performing **kinematic analysis in ADAMS** and applying **Generative Design**, **topology optimization**, and **FEM simulation** to achieve efficient packaging and minimal weight.
- **Chassis Analysis & Manufacturing:** Conducted **Ansys ACP (Composite) analysis** to validate the **carbon fiber chassis** design and actively participated in its **hands-on manufacturing** using the **Resin Infusion method**.
- **Prototyping & Validation:** Prototyped topology-optimized suspension components using **Metal 3D Printing (DMLS)**, then validated them on a custom-built, **quarter-model test bench** to conduct **movement and load testing** against **ADAMS** simulation data.
- **Fuel Cell Test Bench Development:** Engineered, fabricated, and electronically integrated a standalone **fuel cell test bench**, executing operational testing (purging, humidity cycling, short-circuiting) and performance data evaluation.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 09/2025 – 12/2025 | Schwieberdingen

- Developed and integrated a **Simulink** model for an existing power-split operating strategy concept into the EVsim (PHEV) module, **validating its fidelity** against reference Excel data.
- Engineered a new, elaborate operating strategy in a **MATLAB script** for Strong **Fuel Cell** Hybrid EVs (SEVs), based on the initial concept; this new strategy focuses on **safeguarding Fuel Cell degradation** and is parameterized by various MAPs (lookup tables) for different battery capacities and drive cycles.
- Executed **Multi-objective,multi-variable parametric optimization** to generate the required MAPs, specifically targeting **charge sustenance** and **minimal fuel consumption**.
- Developed **post-processing plots** to analyze **efficiency** and **degradation KPIs** , **benchmarking** the new strategy's performance against DP and ECMS.
- Preparing the developed **FCEV strategy** and co-authoring a **scientific paper** for **academic publication**.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric DoorOpen robot, focusing on its **dynamics** and **mechanical reliability** using iterative simulation strategies in Fusion 360.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Door-open Robot** by creating a single source of truth in **Autodesk Fusion 360 Manage**, which integrated 3D CAD models, electronic datasheets, user manuals, FEM/thermal simulation results, and manufacturing drawings.
- Developed and built a custom **test rig** to evaluate the **clamping performance** of a **robot's Chassis** across varied surfaces (steel, aluminum, carbon fiber) to optimize **material selection and reduce weight** .
- Utilized **rapid prototyping (Raise 3D)** for the fabrication and continuous design refinement of over 150 parts.

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- **Pitched and co-developed the initial concept** for a lightweight, foldable electric wheelchair, which featured a user-removable Li-ion battery pack.
- Conducted a **comprehensive analysis** to **select the optimal battery cell**, evaluating critical metrics including energy density, use case,thermal stability, weight, and cycle lifetime.
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- Catia V5
- NX SIEMENS
- FUSION 360
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency

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linkedin.com/in/vallabh-pataneni

Summary

A **highly motivated** Master's student specializing in **Alternative Drives and Vehicle Systems**. I bring strategic expertise across the entire **E-Mobility development lifecycle**, from high-fidelity **FCEV simulation (MATLAB/Simulink)** and **powertrain optimization** to **Techno-Economic (TCO/LCA) analysis**. My profile demonstrates exceptional **leadership** managing complex vehicle projects, combining **Industrial Rigor** learned at Bosch with **full component development, prototyping, and assembly** gained at Ecogenium. Seeking a Final Thesis opportunity at MAHLE to actively contribute to the **Vehicle Systems project house** by researching and evaluating new technology trends.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University

Relevant Coursework:Automotive Dynamics and Components, Alternative Propulsion Systems, Thermodynamics,Mechatronics, Simulation Sciences.

GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity

GPA: 1.3

Academic Project

Well-to-Wheel (WTW) & TCO Analysis of EVs vs. ICEs, 01/2022 – 06/2022
JNTUHCEH

- Conducted an **extensive literature review** on existing WTW analyses, then **modified key parameters** (e.g., energy mix, consumer costs) to create a model for the **Indian market**.
- Performed a comprehensive **Well-to-Wheel (WTW) sustainability analysis** based on this model to critically evaluate regional "zero-emission" claims by assessing local energy generation.
- Executed a **Total Cost of Ownership (TCO)** and **amortization analysis** to compare the lifetime affordability of EVs and ICEs, providing a complete, region-specific economic and environmental picture.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 09/2025 – 12/2025 | Schwieberdingen

- Developed a **Simulink** model for an existing power-split operating strategy concept and integrated it into the FCEVsim (PHEV) simulation module.
- **Validated** the Simulink model by **correlating** its simulation outputs against the original developer's **reference Excel data** to ensure high model fidelity.
- Engineered a new, elaborate operating strategy in a **MATLAB script** for Strong **Fuel Cell** Hybrid EVs (SEVs), based on the initial concept; this new strategy focuses on **safeguarding Fuel Cell degradation** and is parameterized by various MAPs (lookup tables) for different battery capacities and drive cycles.
- Conducted **parametric optimization** to generate the required MAPs, specifically targeting **charge sustenance** and **minimal fuel consumption**.
- Developed **post-processing plots** to analyze **efficiency** and **degradation KPIs** , **benchmarking** the new strategy's performance against DP and ECMS.
- Preparing the developed FCEV strategy for academic publication, conducting an extensive **literature review** and co-authoring the **scientific paper** with supervision.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric robot, focusing on its **dynamics and mechanical reliability** using iterative simulation strategies in Fusion 360.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Door-open Robot** by creating a single source of truth in **Autodesk Fusion 360 Manage**, which integrated 3D CAD models, electronic datasheets, user manuals, FEM/thermal simulation results, and manufacturing drawings.
- Developed and built a custom **test rig** to evaluate the **clamping performance** of a **robotic component** across varied surfaces (steel, aluminum, carbon fiber) to optimize **material selection and reduce weight** .
- Utilized **rapid prototyping (3D printing)** for the fabrication and continuous design refinement of over 150 parts.

Student Club

Ecogenium e.V, 06/2024 – Present | Aachen, Germany

Mechanical Team lead & member of pit crew 24/25

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- **CAD & Mechanical Design:** Led the **CAD design (Siemens NX)** and FEM simulation for the vehicle's **suspension systems**. Utilized **generative design** for **topology optimization** and designed for various manufacturing processes (DFM), including **prototyping components** with **metal 3D printing (DMLS)**.
- **Fuel Cell Test Bench & Analysis:** Developed and prototyped a standalone **fuel cell test bench** to conduct comprehensive **thermodynamic analysis** (e.g., purging, humidity cycling) and data evaluation to optimize powertrain performance.
- **Strategy Implementation:** Developed and implemented **data-driven driving and control strategies** using **Python** and **Simulink**, leveraging competition data to identify performance bottlenecks and improve vehicle efficiency.
- **Hands-on Fabrication & Maintenance:** Managed the complete vehicle build, **gaining hands-on experience** with **Fuel Cell operation** , **Safety protocol** , **on-track troubleshooting**, and **rapid maintenance**.

Skills

CAD-Proficiency
- Catia V5
- NX SIEMENS
- FUSION 360
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Matlab/Simulink
- ANSYS STRUCTURAL & FLEUNT
- DYMOLA
- Solidworks
- Siemens NX

Software-Proficiency
- PYTHON
- C/C++
- MS Office Suit

PLM Software - Proficiency
- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*

A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Academic Recognization, *RWTH*

Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.3 in core automotive engineering courses.

S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering -RWTH Aachen

vallabh.pataneni@rwth-aachen.de   +49 17685944047   Bahnhof Str 59/2, Renningen

linkedin.com/in/vallabh-pataneni

Summary

A highly dedicated Automotive Engineering Master's student, I am focused on making **heavy-duty hydrogen mobility** a commercial reality. My background merges robust **top-down** FCEV systems expertise, demonstrated at Bosch GMBH by developing simulation models and advanced Driving and Operating strategies, with practical **bottom-up** component knowledge. This hands-on experience includes leading the mechanical lifecycle for the Germany's only student-developed Fuel Cell vehicle built from scratch. I bring proven proficiency across **vehicle simulation (MATLAB/Simulink), mechanical design (NX, Catia V5), structural analysis (FEM/CFD), PLM** and **Prototyping**. I am now seeking an **Internship or Master's Thesis** opportunity at **EKPO Fuel Cell Technologies** to transition my expertise and gain direct technical mastery in the development, testing, and analysis of PEM fuel cell stack components, directly aligning my comprehensive skills with EKPO's global technology leadership.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 06/2025 – 12/2025 | Schwieberdingen

- Developed a **MATLAB/Simulink** model for an existing power-split operating strategy concept and integrated it into the FCEVsim (PHEV) simulation module.
- **Validated** the Simulink model by **correlating** its simulation outputs against the original developer's **reference Excel data** to ensure high model fidelity.
- Engineered a new, elaborate operating strategy in a **MATLAB script** for Strong **Fuel Cell** Hybrid EVs (SEVs), based on the initial concept; this new strategy focuses on **safeguarding Fuel Cell degradation** and is parameterized by various MAPs (lookup tables) for different battery capacities and drive cycles.
- Conducted **parametric optimization** to generate the required MAPs, specifically targeting **charge sustenance** and **minimal fuel consumption**.
- Developed **post-processing plots** to analyze **efficiency** and **degradation KPIs** , **benchmarking** the new strategy's performance against DP and ECMS.
- Preparing the developed FCEV strategy for academic publication, conducting an extensive **literature review** and co-authoring the **scientific paper** with supervision.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric robot, focusing on its **dynamics and mechanical reliability** using iterative simulation strategies in Fusion 360.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Door-open Robot** by creating a single source of truth in **Autodesk Fusion 360 Manage**, which integrated 3D CAD models, electronic datasheets, user manuals, FEM/thermal simulation results, and manufacturing drawings.
- Developed and built a custom **test rig** to evaluate the **clamping performance** of a **robotic component** across varied surfaces (steel, aluminum, carbon fiber) to optimize **material selection and reduce weight** .
- Utilized **rapid prototyping (3D printing)** for the fabrication and continuous design refinement of over 150 parts.

Student Club

Ecogenium e.V, *Mechanical Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- **CAD & Mechanical Design:** Led the **CAD design (Siemens NX)** and FEM simulation for the vehicle's **suspension systems**. Utilized **generative design** for **topology optimization** and designed for various manufacturing processes (DFM), including **prototyping components** with **metal 3D printing (DMLS)**.
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- **Strategy Implementation:** Developed and implemented **data-driven driving and control strategies** using **Python** and **Simulink**, leveraging competition data to identify performance bottlenecks and improve vehicle efficiency.
- **Hands-on Fabrication & Maintenance:** Managed the complete vehicle build, **gaining hands-on experience** with **Fuel Cell operation** , **Safety protocol** , **on-track troubleshooting**, and **rapid maintenance**.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework:Automotive Dynamics and Components, Alternative Propulsion Systems, Thermodynamics,Mechatronics, Simulation Sciences.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Academic Project

Electric Wheelchair with Modular Battery System, 10/2024 – 02/2024
ISEA-RWTH

- **Pitched and co-developed the initial concept** for a lightweight, foldable electric wheelchair, which featured a user-removable Li-ion battery pack.
- Conducted a **comprehensive analysis** to **select the optimal battery cell**, evaluating critical metrics including energy density, use case,thermal stability, weight, and cycle lifetime.
- **Engineered the complete battery pack architecture**, including **power electronics** and charging strategy, and developed a foundational **Simulink model** to **simulate discharge curves** and **estimate vehicle range.**

Skills

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- Catia V5
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Summary

A **highly capable** Automotive Engineering Master's student whose passion is entirely dedicated to making **hydrogen mobility** a commercial reality. I bring deep technical understanding of **Fuel Cell operation, degradation, and efficient system utilization** a knowledge base spanning industrial simulation at **Bosch** and **hands-on testing** at Ecogenium e.V. My unique profile merges **Industrial Rigor** with **Applied Component Mastery**, providing proven proficiency in **whole-vehicle simulation (MATLAB/Simulink)**, **thermodynamic analysis**, **core mechanical design (CAD)** and **analysis (FEM,CFD)**, and **custom test bench fabrication**. Seeking this mandatory internship to transition my expertise from FCEV systems optimization to the **direct development, testing, and analysis of PEM fuel cell stack components** at ElringKlinger.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 06/2025 – 12/2025 | Schwieberdingen

- Developed a **MATLAB/Simulink** model for an existing power-split operating strategy concept and integrated it into the FCEVsim (PHEV) simulation module.
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Student Club

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- **CAD & Mechanical Design:** Led the **CAD design (Siemens NX)** and FEM simulation for the vehicle's **suspension systems**. Utilized **generative design** for **topology optimization** and designed for various manufacturing processes (DFM), including **prototyping components** with **metal 3D printing (DMLS)**.
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Electric Wheelchair with Modular Battery System, 10/2024 – 02/2024
ISEA-RWTH

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- **Engineered the complete battery pack architecture**, including **power electronics** and charging strategy, and developed a foundational **Simulink model** to **simulate discharge curves** and **estimate vehicle range.**

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linkedin.com/in/vallabh-pataneni

Summary

An Automotive Engineering Master's student defined by a deep commitment to **hydrogen mobility** and **fuel cell technology**. This passion is proven not just in theory, but through practice -professionally, as a **Simulation Intern at Bosch** developing industrial **FCEV operating strategies**, and personally, by **leading** the technical team at **Ecogenium e.V.** to build the Germany's only **fuel cell Student vehicle** from scratch. This dual 'top-down' industrial and 'bottom-up' hands-on experience provides high proficiency in **whole-vehicle simulation (MATLAB/Simulink, Python)**, **thermodynamic analysis**, and benchmarking performance to generate the critical technical data required for **techno-economic assessments (TCO)**. Seeking this Master Thesis to apply this unique FCEV expertise to the economic viability of new refrigeration systems.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 06/2025 – 12/2025 | Schwieberdingen

- Developed a **MATLAB/Simulink** model for an existing power-split operating strategy concept and integrated it into the EVsim (PHEV) simulation module.
- **Validated** the Simulink model by **correlating** its simulation outputs against the original developer's **reference Excel data** to ensure high model fidelity.
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Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric robot, focusing on its **dynamics and mechanical reliability** using iterative simulation strategies in Fusion 360.
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- Developed and built a custom **test rig** to evaluate the **clamping performance** of a **robotic component** across varied surfaces (steel, aluminum, carbon fiber) to optimize **material selection and reduce weight** .
- Utilized an **industry-standard dual-nozzle FDM** rapid prototyping 3D printer (Raise 3D) for the fabrication and continuous design refinement of over **150 parts** based on physical testing feedback.

Student Club

Ecogenium e.V, *Mechanical Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- **CAD & Mechanical Design:** Led the **CAD design (Catia V5)** and FEM simulation for the vehicle's **suspension systems**. Utilized **generative design** for **topology optimization** and designed for various manufacturing processes (DFM), including **prototyping components** with **metal 3D printing (DMLS)**.
- **Fuel Cell Test Bench & Analysis:** Engineered and programmed a standalone **fuel cell test bench** to conduct comprehensive **thermodynamic analysis** (e.g., purging, humidity cycling) and data evaluation to optimize powertrain performance.
- **Data-Driven Strategy:** Developed and implemented **data-driven driving strategies** using **Python** and **Simulink**, leveraging competition data to identify performance bottlenecks and improve vehicle efficiency.
- **Hands-on Fabrication & Maintenance:** Managed the complete vehicle build, **gaining hands-on experience** with component assembly, **on-track troubleshooting**, and **rapid maintenance**.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework:Automotive Dynamics and Components, Alternative Propulsion Systems, Thermodynamics,Mechatronics, Simulation Sciences.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Academic Project

Well-to-Wheel (WTW) & TCO Analysis of EVs vs. ICEs, 01/2022 – 06/2022
JNTUHCEH

- Conducted an **extensive literature review** on existing WTW analyses, then **modified key parameters** (e.g., energy mix, consumer costs) to create a model for the **Indian market**.
- Performed a comprehensive **Well-to-Wheel (WTW) sustainability analysis** based on this model to critically evaluate regional "zero-emission" claims by assessing local energy generation.
- Executed a **Total Cost of Ownership (TCO)** and **amortization analysis** to compare the lifetime affordability of EVs and ICEs, providing a complete, region-specific economic and environmental picture.

Skills

Simulation-Proficiency
- Matlab/Simulink
- ANSYS STRUCTURAL & FLEUNT
- DYMOLA
- Solidworks
- Siemens NX

CAD-Proficiency
- Catia V5
- NX SIEMENS
- FUSION 360
- PTC CREO
- SOLIDWORKS

Software-Proficiency
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Summary

A proactive and committed Master's student in Automotive Engineering with a strong focus on **component testing, vehicle dynamics, and quality assurance**. Brings professional experience from Bosch in **simulation and data analysis (MATLAB/Simulink)** and extensive hands-on experience as a team lead for a student vehicle club (Ecogenium). Highly proficient in the end-to-end component lifecycle: from **CAD modeling (Catia V5, Siemens NX)** and simulation to **designing and fabricating test benches** and validating performance. This is complemented by direct experience in the **development, testing, and hands-on maintenance of braking systems**. Seeking a Master Thesis to apply this dual expertise to the development of a quality-testing device for BMW Motorrad.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework:Automotive Dynamics and Components, Material Science, Sensor Technology,Mechatronics, Simulation Sciences, DFM, FEM.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

Ecogenium e.V, *Mechanical Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- **CAD & Mechanical Design:** Led the **CAD design (Catia V5)** and FEM simulation for the vehicle's **suspension systems**. Utilized **generative design** for **topology optimization** and designed for various manufacturing processes (DFM), including **prototyping components** with **metal 3D printing (DMLS)**.
- **Braking System Development:** Led the design and prototyping of the vehicle's **hydraulic braking system** as part of the overall suspension and steering development. This included **selecting the system layout, master cylinders, and calipers**, managing full **vehicle integration**, and conducting **performance evaluation, testing, and routine maintenance (e.g., bleeding brakes)**.
- **Test Bench Design & Validation:** Engineered, fabricated, and operated multiple **custom test benches** (for fuel cells, suspension kinematics) to conduct comprehensive experimental analysis and **collect precise measurement data**.
- **Hands-on Fabrication & Repair:** Managed the complete vehicle build, gaining hands-on experience with component assembly, on-track troubleshooting, and **rapid repair** under deadlines.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 06/2025 – 12/2025 | Schwieberdingen

- Developed a **MATLAB/Simulink** model for an existing power-split operating strategy concept and integrated it into the EVsim (PHEV) simulation module.
- **Validated** the Simulink model by **correlating** its simulation outputs against the original developer's **reference Excel data** to ensure high model fidelity.
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- Utilized an **industry-standard dual-nozzle FDM** rapid prototyping 3D printer (Raise 3D) for the fabrication and continuous design refinement of over **150 parts** based on physical testing feedback.

Additive Manufacturing Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- Contributed to the **conceptualization** and **prototyping** of a proof-of-concept air taxi (scaled model) using **SolidWorks**, with a focus on translating mobility ideas into functional designs.
- Developed detailed **CAD models**, conducted **FEM simulations**, and performed **rapid prototyping** for intricate, flight-critical parts including wings and the landing mechanism.
- Conducted comprehensive **FEM and CFD simulations** on the fuselage and various aerofoil designs to optimize their geometry, resulting in reduced weight and enhanced aerodynamic performance.
- Drove **Design for Additive Manufacturing (DfAM)** principles by **strategically splitting** large components to **maximize production bed usage** and integrating **robust fixing mechanisms**, which **reduced print times** and **post-processing efforts**.

Skills

CAD-Proficiency

- Catia V5
- NX SIEMENS
- FUSION 360
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency

- Matlab/Simulink
- ANSYS STRUCTURAL & FLEUNT
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Summary

A **highly motivated** Automotive Engineering Master's student with current professional experience as a **Simulation Intern** at **Robert Bosch GmbH (PS)**. Demonstrates proven capabilities in the **end-to-end simulation lifecycle**, including complex system **modeling, optimization** and **Post-processing in MATLAB/Simulink**, as well as rigorous **model validation against reference data**. Adept at defining **Key Performance Indicators (KPIs)** for performance benchmarking and analyzing **system mechanics**. This core simulation competency is paired with significant hands-on experience in **Vehicle-Component Development** and **Testing** from leading student vehicle projects. Seeking a Master Thesis to leverage this dual expertise in **simulation** and **applied mechanics** for the inertial sensor test facility model.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 06/2025 – 12/2025 | Schwieberdingen

- Developed a **MATLAB/Simulink** model for an existing power-split operating strategy concept and integrated it into the EVsim (PHEV) simulation module.
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- Conducted **parametric optimization** to generate the required MAPs, specifically targeting **charge sustenance** and **minimal fuel consumption**.
- Developed **post-processing plots** to analyze **efficiency** and **degradation KPIs** , **benchmarking** the new strategy's performance against DP and ECMS.
- Future work: **Optimizing** fuel consumption via FC ON/OFF control, developing an SOH model for **degradation evaluation**, and integrating GPS/elevation data for **predictive control**.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric robot, focusing on its **dynamics and mechanical reliability** using iterative simulation strategies in Fusion 360.
- Developed and built a custom **test rig** to evaluate the **clamping performance** of a **robotic component** across varied surfaces (steel, aluminum, carbon fiber) to optimize **material selection and reduce weight** .
- Programmed a **motor controller** using Arduino IDE to implement **precise acceleration** and **deceleration curves**, demonstrating knowledge of **mechanical component** behavior.
- **Developed** and **tested** the **robot noise cancellation** setup by placing microphones at different positions and orientations inside the robot and **evaluating performance** with controlled noise signals to select a robust configuration

Student Club

Ecogenium e.V, *Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- Developing a track-based vehicle suspension model in **Adams CAR** and integrating its data with **Simulink** for co-simulation.
- Conducting **parameter optimization** on suspension coordinates to improve vehicle roll and minimize toe/camber changes during bump and cornering maneuvers.
- Engineered and programmed a standalone **fuel cell test bench** to conduct comprehensive experimental analysis, including data evaluation to optimize powertrain performance and develop operating strategies .
- Developed and implemented **data-driven** driving strategies using **Python, Simulink**, and Power BI, leveraging competition data to identify performance bottlenecks and significantly improve overall vehicle efficiency
- Led the simulation-driven design of the vehicle's aeroshell, conducting **CFD analysis** in **Ansys** to analyze wake dynamics and pressure distribution, optimizing aerodynamic performance.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework:Automotive Dynamics and Components, Material Science, Sensor Technology,Mechatronics, Simulation Sciences, DFM, FEM.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Academic Project

Electric Wheelchair with Modular Battery System, 10/2024 – 02/2024
ISEA-RWTH

- **Pitched and co-developed the initial concept** for a lightweight, foldable electric wheelchair, which featured a user-removable Li-ion battery pack.
- Conducted a **comprehensive analysis** to **select the optimal battery cell**, evaluating critical metrics including energy density, use case,thermal stability, weight, and cycle lifetime.
- **Engineered the complete battery pack architecture**, including **power electronics** and charging strategy, and developed a foundational **Simulink model** to **simulate discharge curves** and **estimate vehicle range.**

Skills

Simulation-Proficiency
- Matlab/Simulink
- ANSYS STRUCTURAL & FLEUNT
- Fusion 360
- Solidworks

Software-Proficiency
- PYTHON
- C/C++
- MS Office Suit

CAD-Proficiency
- NX SIEMENS
- FUSION 360
- Catia V5
- PTC CREO
- SOLIDWORKS

Data Analysis - Proficiency
Power Bi
Tableau

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Academic Recognization, *RWTH*
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.3 in core automotive engineering courses.

S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering - RWTH Aachen

vallabh.pataneni@rwth-aachen.de +49 17685944047 Stuttgart linkedin.com/in/vallabh-pataneni

Summary

A Master's student in Automotive Engineering specializing in **drive system simulation** and **control engineering**. Brings proven **leadership** experience from managing multidisciplinary vehicle projects. My simulation expertise is demonstrated through two key areas: industrial experience at **Robert Bosch GmbH** developing **hybrid (PHEV, FCEV)** operating strategies, and hands-on application at **Ecogenium e.V.**, where I implemented data-driven **MATLAB/Simulink** control strategies that directly improved vehicle efficiency. Highly proficient in **Dymola**, modeling **multi-physical systems**, and analyzing KPIs like **fuel consumption** to support drive development.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 09/2025 – 12/2025 | Schwieberdingen

- Developed and integrated a iterative **Simulink** model for an existing Rule-based power-split operating strategy concept into the EVsim (PHEV) module, **validating its fidelity** against reference Excel data.
- Engineered a new, elaborate operating strategy in a **MATLAB script** for Strong **Fuel Cell** Hybrid EVs (SEVs), based on the initial concept; this new strategy focuses on **safeguarding Fuel Cell degradation** and is parameterized by various MAPs (lookup tables) for different battery capacities and drive cycles.
- Executed **Multi-objective,multi-variable parametric optimization** to generate the required MAPs, specifically targeting **charge sustenance** and **minimal fuel consumption**.
- Developed **post-processing plots** to analyze **efficiency** and **degradation KPIs** , **benchmarking** the new strategy's performance against DP and ECMS.
- Currently researching extensions of this rule-based strategy into a predictive control strategy, incorporating live **GPS** and **road elevation data** to anticipate vehicle power demands.
- **Co-authoring a scientific paper** based on our State of the Art FCEV strategy.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric DoorOpen robot, focusing on its **dynamics** and **mechanical reliability** using iterative simulation strategies in Fusion 360.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Door-open Robot** by creating a single source of truth in **Autodesk Fusion 360 Manage**, which integrated 3D CAD models, electronic datasheets, user manuals, FEM/thermal simulation results, and manufacturing drawings.
- Developed an iterative, data-driven **MATLAB** script to program a motor controller, successfully replicating human-like, user-centric acceleration and deceleration curves. This involved collecting **real-world measurement data** and integrated it to the motor control code.
- Developed and built a custom **test rig** to evaluate the **clamping performance** of a **robot's Chassis** across varied surfaces (steel, aluminum, carbon fiber) to optimize **material selection and reduce weight** .
- Utilized **rapid prototyping (Raise 3D)** for the fabrication and continuous design refinement of over 150 parts.

Student Club

Ecogenium e.V, *Mechanical Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- **Fuel Cell Test Bench Development:**Engineered, fabricated, and electronically integrated **custom-built test benches** (standalone fuel cell, quarter-model suspension) to conduct **performance data evaluation** and validate simulation models against real-world hardware data.
- **Strategy Implementation:** Architected and deployed advanced driving and fuel cell **temperature control strategies** using **Simulink**; leveraged real-time competition data to **boost fuel efficiency by 80%**, a critical factor in securing a **runner-up finish** at the Shell Eco-Marathon.
- **Suspension Development: Engineered** the complete **front (double wishbone)** and **rear (trailing arm) suspension systems** and its **Components** using **Catia V5** and **Siemens NX**, performing **kinematic analysis in ADAMS** and applying **Generative Design**, **topology optimization**, and **FEM simulation** to achieve efficient packaging and minimal weight.
- **Prototyping & Validation:** Prototyped topology-optimized suspension components using **Metal 3D Printing (DMLS)**, then validated them on a custom-built, **quarter-model test bench**.

Education

Master of Science in Automotive Engineering, 09/2023 – Present | Aachen, Germany
RWTH Aachen University
Relevant Coursework:Alternative Propulsion Systems, Thermodynamics, Mechatronics, Simulation Sciences, Material Science, Energy Transition.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
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Academic Project

Electric Wheelchair with Modular Battery System, 10/2024 – 02/2024
ISEA-RWTH

- **Pitched and co-developed the initial concept** for a lightweight, foldable electric wheelchair, which featured a user-removable Li-ion battery pack.
- Conducted a **comprehensive analysis** to **select the optimal battery cell**, evaluating critical metrics including energy density, use case,thermal stability, weight, and cycle lifetime.
- **Engineered the complete battery pack architecture**, including **power electronics** and charging strategy, and developed a foundational **Simulink model** to **simulate discharge curves** and **estimate vehicle range.**

Skills

CAD-Proficiency
- Catia V5
- NX SIEMENS
- FUSION 360
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Matlab/Simulink
- ANSYS STRUCTURAL & FLEUNT
- DYMOLA
- Solidworks
- Siemens NX

Software-Proficiency
- PYTHON
- C/C++
- MS Office Suit

PLM Software - Proficiency
- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Languages

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Fluent (C1)

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Intermediate (B1)

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Summary

A highly motivated Automotive Engineering Master's student specializing in **drivetrain operating strategy** and **vehicle data validation**. I combine professional simulation experience from **Robert Bosch GmbH** developing **operating strategies** for PHEVs and SEVs focusing on consumption, with proven **leadership in conventional vehicle development** from **Ecogenium e.V.**, where I managed a team through the entire build-and-test lifecycle. I am highly proficient in **MATLAB** and **Python** for large **data analysis** and possess a strong, hands-on initiative for practical problem-solving and building test hardware. I am seeking this internship to apply my dual expertise in industrial simulation and practical vehicle engineering to BMW's innovative **PHEV** drive systems.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* — 09/2025 – 12/2025 | Schwieberdingen

- Developed and integrated a iterative **Simulink** model for an existing Rule-based power-split operating strategy concept into the EVsim (PHEV) module, **validating its fidelity** against reference Excel data.
- Engineered a new, elaborate operating strategy in a **MATLAB script** for Strong **Fuel Cell** Hybrid EVs (SEVs), based on the initial concept; this new strategy focuses on **safeguarding Fuel Cell degradation** and is parameterized by various MAPs (lookup tables) for different battery capacities and drive cycles.
- Executed **Multi-objective,multi-variable parametric optimization** to generate the required MAPs, specifically targeting **charge sustenance** and **minimal fuel consumption**.
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- Currently researching extensions of this rule-based strategy into a predictive control strategy, incorporating live **GPS** and **road elevation data** to anticipate vehicle power demands.
- **Co-authoring a scientific paper** based on our State of the Art FCEV strategy.

Industrial Design Intern, *Techolution* — 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric DoorOpen robot, focusing on its **dynamics** and **mechanical reliability** using iterative simulation strategies in Fusion 360.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Door-open Robot** by creating a single source of truth in **Autodesk Fusion 360 Manage**, which integrated 3D CAD models, electronic datasheets, user manuals, FEM/thermal simulation results, and manufacturing drawings.
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Student Club

Ecogenium e.V, *Mechanical Team lead & member of pit crew 24/25* — 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
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- **Prototyping & Validation:** Prototyped topology-optimized suspension components using **Metal 3D Printing (DMLS)**, then validated them on a custom-built, **quarter-model test bench**.

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GPA: 1.7

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Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Academic Project

Electric Wheelchair with Modular Battery System, 10/2024 – 02/2024
ISEA-RWTH

- **Pitched and co-developed the initial concept** for a lightweight, foldable electric wheelchair, which featured a user-removable Li-ion battery pack.
- Conducted a **comprehensive analysis** to **select the optimal battery cell**, evaluating critical metrics including energy density, use case,thermal stability, weight, and cycle lifetime.
- **Engineered the complete battery pack architecture**, including **power electronics** and charging strategy, and developed a foundational **Simulink model** to **simulate discharge curves** and **estimate vehicle range.**

Skills

CAD-Proficiency
- Catia V5
- NX SIEMENS
- FUSION 360
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
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Summary

A highly motivated Automotive Engineering Master's student with a career focus on making **Hydrogen mobility** a commercial reality. My experience provides a unique, holistic perspective, combining **'top-down' system-level engineering** with **'bottom-up' core-level operational Engineering**. At **Robert Bosch GmbH**, I am developing **FCEV operating strategies** for PHEVs and SEVs focusing on **fuel consumptions** and **degradation** using **MATLAB**. Concurrently, at **Ecogenium e.V.**, I gained **operational expertise** by leading the hands-on development of a **hydrogen fuel cell vehicle**, where I contributed to the **complete component lifecycle** (CAD, simulation, validation) and have gained deep, practical expertise in **fuel cell operation, testing, and the implementation of data-driven control strategies (MATLAB/Simulink)** that resulted in direct **efficiency improvement**. I am eager to apply my dual system-strategy and applied operational skills to this internship, seeing it as the perfect bridge to gain the **'complete package'** of FCEV expertise by contributing to your component data analysis goals.

Education

Master of Science in Automotive Engineering, 09/2023 – Present | Aachen, Germany

RWTH Aachen University

Relevant Coursework:Alternative Propulsion Systems, Thermodynamics, Mechatronics, Simulation Sciences, Material Science, Energy Transition.

GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India

Jawaharlal Nehru Technological Univerisity

GPA: 1.3

Student Club

Ecogenium e.V, *Mechanical Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- **Fuel Cell Test Bench Development:** Engineered, fabricated, and electronically integrated a standalone **fuel cell test bench**, executing operational testing (purging, humidity cycling, short-circuiting) and performance data evaluation.
- **Strategy Implementation:** Architected and deployed advanced driving and fuel cell **temperature control strategies** using **Simulink**; leveraged real-time competition data to **boost fuel efficiency by 80%**, a critical factor in securing a **runner-up finish** at the Shell Eco-Marathon.
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- **Prototyping & Validation:** Prototyped topology-optimized suspension components using **Metal 3D Printing (DMLS)**, then validated them on a custom-built, **quarter-model test bench**.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 09/2025 – 12/2025 | Schwieberdingen

- Developed and integrated a iterative **Simulink** model for an existing power-split operating strategy concept into the EVsim (PHEV) module, **validating its fidelity** against reference Excel data.
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- Executed **Multi-objective,multi-variable parametric optimization** to generate the required MAPs, specifically targeting **charge sustenance** and **minimal fuel consumption**.
- Developed **post-processing plots** to analyze **efficiency** and **degradation KPIs** , **benchmarking** the new strategy's performance against DP and ECMS.
- Currently **co-authoring a scientific paper** based on our State of the Art FCEV strategy.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric DoorOpen robot, focusing on its **dynamics** and **mechanical reliability** using iterative simulation strategies in Fusion 360.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Door-open Robot** by creating a single source of truth in **Autodesk Fusion 360 Manage**, which integrated 3D CAD models, electronic datasheets, user manuals, FEM/thermal simulation results, and manufacturing drawings.
- Developed and built a custom **test rig** to evaluate the **clamping performance** of a **robot's Chassis** across varied surfaces (steel, aluminum, carbon fiber) to optimize **material selection and reduce weight** .
- Utilized **rapid prototyping (Raise 3D)** for the fabrication and continuous design refinement of over 150 parts.

Academic Project

Electric Wheelchair with Modular Battery System, 10/2024 – 02/2024
ISEA-RWTH

- **Pitched and co-developed the initial concept** for a lightweight, foldable electric wheelchair, which featured a user-removable Li-ion battery pack.
- Conducted a **comprehensive analysis** to **select the optimal battery cell**, evaluating critical metrics including energy density, use case,thermal stability, weight, and cycle lifetime.
- **Engineered the complete battery pack architecture**, including **power electronics** and charging strategy, and developed a foundational **Simulink model** to **simulate discharge curves** and **estimate vehicle range.**

Skills

CAD-Proficiency
- Catia V5
- NX SIEMENS
- FUSION 360
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Matlab/Simulink
- ANSYS STRUCTURAL & FLEUNT
- DYMOLA
- Solidworks
- Siemens NX

Software-Proficiency
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PLM Software - Proficiency
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- Siemens Teamcentre

Languages

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Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
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vallabh.pataneni@rwth-aachen.de +49 17685944047 Bahnhof Str 59/2, Renningen

linkedin.com/in/vallabh-pataneni

Summary

A proactive and committed Automotive Engineering Master's student with a strong focus on **chassis development**, **vehicle dynamics**, and **component simulation**. Brings professional experience from Bosch in **MATLAB/Simulink simulation and data analysis** combined with extensive **hands-on experience** leading a student vehicle development team (Ecogenium e.V.). Highly proficient in the end-to-end component lifecycle: from **CAD modeling** (**Catia V5**, **Siemens NX**) and **FEM simulation** to **topology optimization**, **test-bench-based performance validation**, and **prototyping** using **advanced manufacturing technologies**. Seeking a mandatory internship to apply this **dual expertise in simulation and hands-on chassis design** to the development of **innovative chassis components at BMW Group**.

Education

Master of Science in Automotive Engineering, 09/2023 – Present | Aachen, Germany

RWTH Aachen University

Relevant Coursework:Automotive Dynamics and Components, Lightweight Design and Optimization, Material Science, Mechatronics, Simulation Sciences, FEM.

GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India

Jawaharlal Nehru Technological Univerisity

GPA: 1.3

Student Club

Ecogenium e.V, *Mechanical Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- **Suspension Development: Engineered** the complete **front (double wishbone)** and **rear (trailing arm) suspension systems** and its **Components** using **Catia V5** and **Siemens NX**, performing **kinematic analysis in ADAMS** and applying **Generative Design**, **topology optimization**, and **FEM simulation** to achieve efficient packaging and minimal weight.
- **Chassis Analysis & Manufacturing:** Conducted **Ansys ACP (Composite) analysis** to validate the **carbon fiber chassis** design and actively participated in its **hands-on manufacturing** using the **Resin Infusion method**.
- **Prototyping & Validation:** Prototyped topology-optimized suspension components using **Metal 3D Printing (DMLS)**, then validated them on a custom-built, **quarter-model test bench** to conduct **movement and load testing** against **ADAMS** simulation data.
- **Strategy Implementation:** Developed and implemented **data-driven driving and control strategies** using **Python** and **Simulink**, leveraging competition data to identify performance bottlenecks and improve vehicle efficiency.

Experience

Operating Strategy - Simulation Intern, *Robert Bosch GMBH* 09/2025 – 12/2025 | Schwieberdingen

- Developed and integrated a **Simulink** model for an existing power-split operating strategy concept into the EVsim (PHEV) module, **validating its fidelity** against reference Excel data.
- Engineered a new, elaborate operating strategy in a **MATLAB script** for Strong **Fuel Cell** Hybrid EVs (SEVs), based on the initial concept; this new strategy focuses on **safeguarding Fuel Cell degradation** and is parameterized by various MAPs (lookup tables) for different battery capacities and drive cycles.
- Executed **Multi-objective,multi-variable parametric optimization** to generate the required MAPs, specifically targeting **charge sustenance** and **minimal fuel consumption**.
- Developed **post-processing plots** to analyze **efficiency** and **degradation KPIs** , **benchmarking** the new strategy's performance against DP and ECMS.
- Preparing the developed FCEV strategy for academic publication, conducting an extensive **literature review** and co-authoring the **scientific paper** with supervision.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Designed, **simulated**, prototyped, and rigorously tested a user-centric DoorOpen robot, focusing on its **dynamics** and **mechanical reliability** using iterative simulation strategies in Fusion 360.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Door-open Robot** by creating a single source of truth in **Autodesk Fusion 360 Manage**, which integrated 3D CAD models, electronic datasheets, user manuals, FEM/thermal simulation results, and manufacturing drawings.
- Developed and built a custom **test rig** to evaluate the **clamping performance** of a **robot's Chassis** across varied surfaces (steel, aluminum, carbon fiber) to optimize **material selection and reduce weight** .
- Utilized **rapid prototyping (Raise 3D)** for the fabrication and continuous design refinement of over 150 parts.

Additive Manufacturing Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- Contributed to the **conceptualization** and **prototyping** of a proof-of-concept air taxi (scaled model) using **SolidWorks**, with a focus on translating mobility ideas into functional designs.
- Developed detailed **CAD models**, conducted **FEM simulations**, and performed **rapid prototyping** for intricate, flight-critical parts including wings and the landing mechanism.
- Conducted comprehensive **FEM and CFD simulations** on the fuselage and various aerofoil designs to optimize their geometry, resulting in reduced weight and enhanced aerodynamic performance.
- Drove **Design for Additive Manufacturing (DfAM)** principles by strategically splitting large components to maximize production bed usage, which reduced print times and post-processing efforts.

Academic Project

Electric Wheelchair with Modular Battery System, 10/2024 – 02/2024
ISEA-RWTH

- **Pitched and co-developed the initial concept** for a lightweight, foldable electric wheelchair, which featured a user-removable Li-ion battery pack.
- Conducted a **comprehensive analysis** to **select the optimal battery cell**, evaluating critical metrics including energy density, use case,thermal stability, weight, and cycle lifetime.
- **Engineered the complete battery pack architecture**, including **power electronics** and charging strategy, and developed a foundational **Simulink model** to **simulate discharge curves** and **estimate vehicle range.**

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Simulation-Proficiency
- Matlab/Simulink
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linkedin.com/in/vallabh-pataneni

Summary

A proactive and committed Master's student in **Automotive Engineering** with a strong focus on vehicle dynamics, component testing, and **damage analysis**. Possesses hands-on experience in the development and maintenance of **braking systems**, including performance evaluation and routine checks . Highly proficient in designing and operating **test benches**, analyzing **test data**, and **correlating simulation results with real-world performance** using tools like MATLAB/Simulink and Python. Seeking a Mandatory internship to apply a deep understanding of **failure mechanisms** in mechanical components and contribute to the development of next-generation braking systems.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework:Automotive Dynamics and Components, Material Science, Mechatronics, Fundamentals of Machine Learning, DFM, FEM.

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

Ecogenium e.V, *Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- Contributed to the development and implementation of the vehicle’s **braking system**, encompassing **testing, evaluation**, and routine maintenance activities like **bleeding brakes** to ensure optimal performance and safety .
- Directed on-track troubleshooting, performing **damage checks** and rapid repairs under competition deadlines to maintain vehicle integrity .
- Engineered and programmed a standalone **fuel cell test bench** to conduct comprehensive experimental analysis, including data evaluation to optimize powertrain performance and develop operating strategies .
- Developed and implemented **data-driven** driving strategies using **Python, Simulink**, and Power BI, leveraging competition data to identify performance bottlenecks and significantly improve overall vehicle efficiency

Experience

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- Developed and built a custom **test rig** to evaluate the **clamping performance** of a **robotic component** across varied surfaces (steel, aluminum, carbon fiber) to optimize **material selection and reduce weight** .
- Conducted **routine maintenance** on **gearboxes** and other moving components to ensure **optimal performance** and **longevity** .
- Programmed a **motor controller** using Arduino IDE to implement **precise acceleration** and **deceleration curves**, demonstrating knowledge of **mechanical component** behavior.
- Utilized an industry standard **dual-nozzle FDM rapid prototyping 3D printer** (Raise 3D) for fabrication and continuous design refinement of over **150** different parts based on testing feedback.

Additive Manufacturing Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- **Designed** and **simulated** various **thermal sinks** to **reduce heat** generated by motors, demonstrating practical application in **thermal management system engineering** for **electric drive subcomponents**.
- **Designed**, **simulated**, **prototyped**, and **rigorously tested** a user-centric door-opening robot, focusing on its **dynamics** and **mechanical reliability** using **iterative design** and **simulation strategies** in Fusion 360.
- Drove **Design for Additive Manufacturing (DfAM)** principles by **strategically splitting** large components to **maximize production bed usage** and integrating **robust fixing mechanisms**, which **reduced print times** and **post-processing efforts**.

Academic Project

Vehicle Sensor Data analysis Project - IKA, 04/2025 – 07/2025

Autonomous Engineer

- Designed and built **Python data pipelines** to process **camera, LiDAR, and radar sensor data**, using **signal processing** and **feature extraction** for **environment mapping** and behavior analysis.
- Developed sensor-driven **testbenches** to validate vehicle control functions, directly linking lab validation with practical outcomes.
- Developed models and tuned controllers using **MATLAB/Simulink and Python**, independently **correlating simulation and real-world results** to optimize vehicle performance through **data-driven analysis** .
- Executed digital simulations and **real-time testing** using a mock vehicle and platforms like CARLA, advancing control system effectiveness.

Skills

CAD-Proficiency

- NX SIEMENS
- FUSION 360
- Catia V5
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency

- Star-CCM+
- ANSYS STRUCTURAL & FLEUNT
- Fusion 360
- Matlab/Simulink
- Solidworks

Software-Proficiency

- PYTHON
- C/C++
- MS Office Suit

PLM Software - Proficiency

- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Data Analysis - Proficiency

Power Bi
Tableau

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Academic Recognization, *RWTH*
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

S V V SAI SRI VALLABH PATANENI

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vallabh.pataneni@rwth-aachen.de | +49 17685944047 | Bahnhof Str 59/2, Renningen

linkedin.com/in/vallabh-pataneni

Summary

Highly motivated automotive engineering master's student from RWTH Aachen University specializing in **lightweight construction**, **vehicle dynamics**, and **simulation**. My experience is highlighted by my **leadership** of a vehicle development team for global competitions, where I guided the project through the entire **design**, **simulation**, and **prototyping** lifecycle. This hands-on role has provided me with a deep understanding of **weight management** principles, from deriving initial **weight targets** to fabricating complex components using advanced manufacturing techniques like **DMLS** and **Resin Infusion**. Currently seeking a **mandatory internship**, I am enthusiastic to contribute these proven skills to a team dedicated to advancing the environmental compatibility of Mercedes-Benz's premium and luxury vehicles through proactive **lightweighting strategies**.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany

RWTH Aachen University

Relevant Coursework:Lightweight Design and Optimization, Material Science, Design for Manufacturing (DFM), Automotive Engineering Dynamics, Components and Assistant Systems, Additive Manufacturing.

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India

Jawaharlal Nehru Technological Univerisity

GPA: 1.3

Student Club

Ecogenium e.V, *Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- Engineered the complete **front (double wishbone)** and **rear (trailing arm) suspension** systems by applying **Generative Design**, **topology optimization**, and **FEM simulation**, creating lightweight components manufactured via **DMLS metal 3D printing** to achieve optimal kinematics and efficient packaging.
- Directed the structural design and **Ansys Composite PrepPost** (ACP) Simulation of Katara's Carbon fibre chassis, utilizing prepreg carbon fiber sandwich layups to achieve a **25 kg mass reduction** while maintaining structural integrity.
- Spearheaded the simulation-driven design of the vehicle’s carbon fiber aeroshell, achieving a **12% reduction in overall drag coefficient**, with these simulation-based gains being **critically validated and proven** through **extensive wind tunnel testing.**
- Oversaw the entire **Product Lifecycle Management (PLM)** process for the vehicle’s core mechanical systems using **Siemens Teamcenter**, managing the Bill of Materials (BOM) and coordinating with planning departments and external suppliers to ensure timeline adherence and seamless integration.

Experience

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- **Designed**, **Simulated** and **Prototyped** a complex user-centric **robot**, focusing on robust mechanical design, systems integration, and applying **Design for Manufacturing and Assembly (DFAM)** principles to ensure ease of maintenance and streamlined production.
- Developed and operated custom test benches to validate the **clampability** of various materials, including aluminum, carbon fiber, and steel, providing crucial data to **optimize component design for weight**, durability, and system integration.
- Established a **unified digital thread** to integrate and manage all 3D CAD models, simulation results, and manufacturing data for a complex mechatronic system, a structured approach with direct application to **strategic vehicle weight management**.
- Oversaw the **rapid prototyping** of over **150 functional components** using an industry-standard Raise 3D printer to support iterative design and testing

Additive Manufacturing Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- Executed the end-to-end **design** and **fabrication** of a proof-of-concept **air taxi model**, performing **CFD and FEM** simulations to validate the **structural integrity** of flight-critical components like **wings** and **landing gear** .
- Executed extensive CFD simulations on the fuselage and aerofoil designs using STAR-CCM+, systematically studying pressure distribution and drag to **refine performance and optimize trade-offs between stability and weight.**
- Drove **Design for Additive Manufacturing (DfAM)** principles by strategically splitting large components to maximize production bed usage and integrating robust fixing mechanisms, which reduced print times and post-processing efforts

Skills

CAD-Proficiency
- NX SIEMENS
- FUSION 360
- Catia V5
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Star-CCM+
- ANSYS STRUCTURAL & FLEUNT
- Fusion 360
- Matlab/Simulink
- Solidworks

Software-Proficiency
- PYTHON
- C/C++
- MS Office Suit

PLM Software - Proficiency
- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Academic Recognization, *RWTH*
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

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vallabh.pataneni@rwth-aachen.de   +49 17685944047   Bahnhof Str 59/2, Renningen

linkedin.com/in/vallabh-pataneni

Summary

A detail-oriented Automotive Engineering Master's student at RWTH Aachen University with a strong foundation in production engineering and manufacturing processes. My experience includes **leading vehicle development teams** for **global competitions** like the Shell Eco-Marathon, which has provided practical skills in **TIG welding**, **structural adhesive bonding**, and project collaboration from concept to execution. I have experience applying analytical tools to enhance process efficiency, including the development of **digital twins** and performing **KPI analysis**. Seeking a **mandatory internship** to apply my academic knowledge and hands-on skills in a highly automated production environment and contribute to challenging industrial projects.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework: Production Engineering, Industrial Engineering, Mechatronics, Automotive Components & Systems, Design for Manufacturing (DFM), Additive Manufacturing, Machine Learning.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

Ecogenium e.V, *Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle, fostering strong teamwork and project management skills in a high-pressure environment.
- Gained foundational experience in metal joining through **TIG welding** on the aluminum-carbon fibre Hybrid chassis and developed expertise in **structural adhesive bonding** for multi-material applications (aluminum-to-carbon fiber), directly relevant to modern body-in-white production.
- Performed the complete **design, Kinematic & FEM simulation** for both **front (double wishbone)** and **rear (trailing arm) suspension** systems for the "Katara" vehicle, specifically aiming for **minimal weight**, **optimal kinematic** characteristics for stability and maneuverability, and efficient packaging within the Eco-Marathon vehicle's constraints.
- Oversaw the entire **Product Lifecycle Management (PLM)** process for the vehicle's core mechanical systems using **Siemens Teamcenter**, managing the Bill of Materials (BOM) and coordinating with planning departments and external suppliers to ensure timeline adherence and seamless integration.
- Produced comprehensive **technical documentation** and led **milestone reviews** with **sponsors**, effectively communicating complex engineering **designs** and **manufacturing processes** to **technical** and **non-technical** stakeholders.

Experience

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- **Designed**, **Simulated** and **Prototyped** a complex user-centric **robot**, focusing on robust mechanical design, systems integration, and applying **Design for Manufacturing and Assembly (DFAM)** principles to ensure ease of maintenance and streamlined production.
- Engineered and performed **virtual commissioning** of a robotic screw-picking assembly cell by creating a complete **digital twin** in CoppeliaSim. This involved validating control logic and optimizing the system for a factory automation task using AI .
- Conducted detailed **KPI analysis** (e.g., cycle time, throughput, equipment utilization) on the simulated production cell to identify bottlenecks and refine the robotic workflow, leading to a projected 15% increase in operational efficiency.
- Engineered and operated **custom test benches** to validate component **durability** and **system integration**, providing crucial data for iterative design refinement and ensuring mechanical reliability under operational loads.

Additive Manufacturing Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- Led **initiatives** to enhance the **efficiency** of additive manufacturing processes across **FDM and SLA platforms**, optimizing **slicing strategies** and **production parameters** to improve part **quality**, increase **output speed**, and reduce **support material waste** .
- Executed the end-to-end **design** and **fabrication** of a proof-of-concept **air taxi model**, performing **CFD and FEM** simulations to validate the **structural integrity** of flight-critical components like **wings** and **landing gear** .
- Gained hands-on fabrication experience by **welding steel structural components** for the vehicle's fuselage and bonding 3D-printed parts to assemble the complete wing and tail sections, reinforcing multi-process manufacturing skills .
- Drove **Design for Additive Manufacturing (DfAM)** principles by strategically splitting large components to maximize production bed usage and integrating robust fixing mechanisms, which reduced print times and post-processing efforts

Skills

CAD-Proficiency

- NX SIEMENS
- FUSION 360
- Catia V5
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency

- Star-CCM+
- ANSYS STRUCTURAL & FLEUNT
- Fusion 360
- Matlab/Simulink
- Solidworks

Production and Manufacturing

KPI Analysis
Production optimization
Welding (TIG, MIG, Resistive Spot)
Adhesive Bonding

PLM Software - Proficiency

- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*

A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Academic Recognization, *RWTH*

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linkedin.com/in/vallabh-pataneni

Summary

A highly motivated Automotive Engineering Master's student at RWTH Aachen University with a specialization in **simulation and process development** for future vehicle technologies. Seeking a mandatory internship at Mercedes-Benz to apply extensive experience in **CFD analysis**, practical manufacturing processes including **welding and bonding**, and **end-to-end component development** to the electrified powertrain team. Proven expertise in using simulation tools like **Star-CCM+** and leading **hands-on fabrication projects** from CAD design to final assembly. Eager to contribute to **innovative manufacturing processes** for powertrain components and drive the **future of electromobility**.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University

Relevant Coursework: CFD, Thermodynamics, Mechatronics, Automotive Components and Systems, Additive Manufacturing, Mobile Propulsion Systems, DFM, FEM.

GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity

GPA: 1.3

Student Club

Ecogenium e.V, *Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle.
- Led the **simulation-driven design** of "Katara's" aeroshell using **Star-CCM+** to refine surface contours and component alignment, achieving a **12% drag reduction**, and **validated** these aerodynamic predictions with **experimental data** from **wind tunnel testing** on a scaled model.
- Analyzed **wake dynamics**, **flow separation**, and **pressure distribution** to provide actionable recommendations that enhanced **airflow efficiency** and resulted in measurable improvements in competition **rollout energy efficiency**.
- Contributed to the manufacturing of an **aluminum-carbon fiber hybrid chassis**, gaining practical experience with **TIG welding**, **aluminum-to-carbon fiber bonding**, and **carbon fiber-to-carbon fiber bonding** processes.
- Oversaw the complete development of a **carbon fiber steering wheel**, from initial **CAD design** and **3D printing** of a test part to creating a polyurethane mold and manufacturing the final component using the **Resin-Infusion method**.

Academic Project

CFD Simulation of Structural Foam Injection for Automotive Manufacturing, *IKV - RWTH*

- Developed a **3D CFD model** in **ANSYS Fluent** to simulate the **injection and expansion** of **polyurethane foam** within a **simplified car's B-pillar**, specifically analyzing the **thermomechanical behavior** during the manufacturing process.
- Conducted **multiphase flow** and **heat transfer analysis** to predict **fill patterns**, identify **potential voids**, and evaluate the **final density distribution** of the cured foam.
- **Contributed to the validation** of the **simulation methodology** against **published experimental data** to create a framework for **optimizing injection parameters** for **structural reinforcement** and **NVH applications**.

Experience

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- **Designed, simulated, prototyped,** and rigorously tested a **user-centric door-opening robot**, focusing on its **dynamics and mechanical reliability** using iterative design and simulation strategies in **Fusion 360**.
- Conducted detailed **Thermal CFD analysis for motor cooling systems and electronics**, simulating various heat sink designs to manage thermal loads and ensure optimal operating temperatures.
- For the **Dooropen Robot** project, managed the complete **digital to physical workflow** by translating final **CAD designs** into manufacturing ready data (**sliced STL files**) and oversaw the **rapid prototyping** of over 150 functional components using Raise 3D printer, bridging the gap between **digital engineering** and physical implementation.

Additive Manufacturing Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- Contributed to the **conceptualization and prototyping** of a **proof-of-concept air taxi** (scaled-down model) using **SolidWorks**, focusing on translating innovative mobility solutions into **tangible designs**.
- Executed **CFD simulations** on the fuselage using **STAR CCM+**, systematically studying **pressure distribution, drag, and lift** while **iterating with multiple aerofoil designs** to refine performance, evaluating and **optimizing trade-offs** between stability, control, and efficiency.
- Performed the **design, simulation, and fabrication** of critical air taxi components (e.g., wings, landing mechanism), specifically **optimizing for maneuverability and structural integrity**.
- Gained **hands-on fabrication experience** by **welding steel structural components** for the fuselage and wings, and **bonding split 3D-printed parts** to assemble the complete wing and tail sections.

Skills

CAD-Proficiency
- NX SIEMENS
- FUSION 360
- Catia V5
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Star-CCM+
- ANSYS STRUCTURAL & FLEUNT
- Fusion 360
- Matlab/Simulink
- Solidworks

PLM Software - Proficiency
- Autodesk Fusion 360 Manage
- Siemens Teamcentre

AM Technology- Proficiency
- FDM
- SLA
- DMLS

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*

A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Academic Recognization, *RWTH*

Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

Bachelor's Thesis, *JNTUH-CEH*

CFD-Based Aerodynamic and **Spray System Design** for Enhanced Performance in **Agricultural Drones**

- Spearheaded aerofoil design with iterative **CFD analysis in ANSYS Fluent**, optimizing drone blade **lift** and **controllability**.
- Engineered and simulated a custom fertilizer spray nozzle using **Ansys Fluent**, performing internal/external flow simulations,methodology transferable to in-cylinder CFD for powertrain applications.

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Master's student in Automotive Engineering -RWTH Aachen

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Summary

An innovative Automotive Engineering Master's student with proven experience in the end-to-end **additive manufacturing** lifecycle, from **generative design** and **parameter studies** to hands-on operation of industrial **DMLS** and **FDM** systems. Combines a strong academic foundation in **material science** and **DFM** with practical experience in **experimental research** and **component characterization**. Seeking a **mandatory internship** to apply advanced skills in workflow optimization and experimental testing to contribute to Fraunhofer IGCV's pioneering research in additive manufacturing.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems), Addtive Manufacturing, Material Science, DFM, DFAM, FEM.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

Ecogenium e.V, *Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle.
- Spearheaded the creation of **topology-optimized** suspension components using **generative design**, orchestrating the full **additive manufacturing workflow** from strategic slicing for the **DMLS process** to post-production and sand blasting.
- Independently operated and supervised the production workflow on **multiple AM systems**, including **SLA, FDM** machines for plastic components and an **advanced robotic arm printer** with a pellet extruder system for large-format moulds.
- Managed the entire **CAD workflow** for **suspension** and **steering** assemblies in **CATIA V5**, ensuring model accuracy and coordinating with chassis and aeroshell teams for iterative integration.
- Orchestrated the end-to-end **experimental validation** process by designing, constructing, and operating bespoke **test rigs** for critical vehicle systems, including a **fuel cell test rig**, a **suspension test bench**, and a **motor testing setup**. Performed systematic **data acquisition and analysis** to **characterize system performance**, validate **structural durability**, and ensure all components met rigorous competition standards.

Experience

Additive Manufacturing Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- Engineered key mechanical systems for a novel, high-speed **FDM 3D printer** specifically designed to process **carbon fiber-infused filaments**, focusing on optimizing the **extruder motion control** and bed heating machinery for mass production.
- Lead the end-to-end design and fabrication of a **proof-of-concept air taxi model**, translating advanced mobility concepts into **functional prototypes** and optimizing complex components like wings and landing gear for efficient **additive manufacturing**.
- Drove **Design for Additive Manufacturing (DfAM)** principles by strategically splitting large Fuselage and Wing compartments of an Air-Taxi to **maximize print bed usage**, integrating robust fixing mechanisms, and performing extensive **post-processing** to achieve a superior surface finish.
- Retrofitted a standard FDM printer for **experimental material research**, designing and integrating a custom **syringe extrusion system** to enable the **additive manufacturing** of non-standard materials like **bio-inks and ceramic pastes**.
- Provided expert printing solutions to clients across five different **FDM and SLA platforms**, simultaneously leading initiatives to enhance **print quality**, speed, and overall **process efficiency**.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- **Designed, simulated, prototyped,** and rigorously tested a **user-centric door-opening robot**, focusing on its **dynamics and mechanical reliability** using iterative design and simulation strategies in **Fusion 360**.
- **Engineered** and **printed** a **specialized test** environment to train an AI model for **robotic screw picking**, developing and printing a fully enclosed test bench, a highly maneuverable camera stand, and a 3-point robotic arm to optimize the machine learning process.
- Developed and operated **custom test benches** to validate **Clampability** of various clamping materials like Aluminium, carbon fibren steel in order to **optimize weight**, **durability**, and **system integration**, providing crucial data for iterative design refinement
- For the **Dooropen Robot** project, managed the complete **digital to physical workflow** by translating final **CAD designs** into manufacturing ready data (**sliced STL files**) and oversaw the **rapid prototyping** of over 150 functional components using Raise 3D printer, bridging the gap between **digital engineering** and physical implementation.

Academic Project

Characterization of Anisotropic Properties in FDM Components, *IKV-RWTH*

- Developed and executed a formal **test plan** to investigate the anisotropic mechanical behavior of FDM-printed parts, directly supporting the need for **parameter study development**.
- Systematically **tested** components **print**ed in **different orientations** using a **three-point** bend test to quantify their relative **flexural strengths**.
- Demonstrated that parts **printed flat** on the build plate exhibited **significantly higher strength**, providing key data for design optimization

Skills

CAD-Proficiency
- NX SIEMENS
- FUSION 360
- Catia V5
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Fusion 360
- Matlab/Simulink
- ANSYS STRUCTURAL & FLEUNT
- CoppeliaSim
- Solidworks

PLM Software - Proficiency
- Autodesk Fusion 360 Manage
- Siemens Teamcentre

AM Technology- Proficiency
- FDM
- SLA
- DMLS
- Bio
- Ceramic

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

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Summary

A highly motivated Automotive Engineering Master's student with a proven track record in the **complete lifecycle** of complex mechanical systems, from initial concept and **design to simulation, prototyping, and hands-on manufacturing** . I am seeking a **mandatory internship** to apply my expertise in **leading development teams** and my passion for **robotics** to the innovative field of **exoskeletons** at Auxsys. My experience includes the **construction of digital twins** for system validation and a keen interest in developing **advanced robotic assistance systems**, with a long-term aim to create **fully autonomous robots** . My skills in **generative design**, **topology optimization**, and building **user-centric robots** align perfectly with the challenges at the forefront of this technology.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems), Mechatronics, DFM, FEM, Machine Learning, Addtive Manufacturing.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

Ecogenium e.V, *Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle.
- Performed the full **CAD design**, **kinematic analysis**, and **FEM simulation** along with the manufacturing for the front **double wishbone** and rear **trailing arm suspension systems** of the "**Katara**" vehicle. Employed **Generative Design** and **topology optimization** methods to achieve **minimal weight** and **enhanced kinematic performance**, utilizing software tools **Siemens NX** and **Fusion 360**.
- Oversaw the entire **Product Lifecycle Management (PLM)** process for the **“Katara”** race car's **suspension and steering systems** leveraging **Siemens Teamcenter**. I established a centralized repository for all engineering assets, spanning from preliminary **CAD** and **simulation models** to the finalized **Production Drawings**, while coordinating **Bill of Materials (BOM)** acquisition with manufacturing collaborators.
- Designed and tested compact **fuel cell rig**, focusing on optimizing the **packaging density** and integration of the **fuel cell** and its associated **electronic systems**.
- Executed the CAD-to-manufacturing process for the **carbon fiber steering wheel**, overseeing its production using the **resin infusion method** and optimizing the composite layup for a **high stiffness-to-weight ratio.**

Experience

Prototyping Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- Contributed to the **conceptualization** and **prototyping** of a **proof-of-concept air taxi (scaled model)** using **SolidWorks**, with a focus on translating advanced mobility ideas into **functional designs**.
- Developed detailed **CAD models**, conducted **FEM simulations**, and performed **rapid prototyping** for intricate, flight-critical parts including **wings** and the **landing mechanism** and achieved an optimal balance between **structural integrity** and **lightweight design** by employing iterative simulation techniques.
- Conducted comprehensive **FEM** and **CFD simulations** on the **fuselage** and various **aerofoil designs** to optimize their geometry, resulting in **reduced weight**, **enhanced lift**, and **minimized drag**.
- Drove **Design for Additive Manufacturing (DfAM)** principles by **strategically splitting large Fuselage** and **Wing compartments of Air-Taxi** to maximize print bed usage, integrating robust **fixing mechanisms**, and performing extensive **post-processing** to achieve a **superior surface finish**.

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- **Designed, simulated, prototyped,** and rigorously tested a **user-centric door-opening robot**, focusing on its **dynamics and mechanical reliability** using iterative design and simulation strategies in **Fusion 360**.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Dooropen Robot**, a complex **mechatronic system**, using **Autodesk Fusion 360 Manage** to create a **single source of truth** integrating multi disciplinary data including **3D CAD models**, electronic Data sheets and User manuals,**FEM/thermal** simulation results and **Manufacturing Drawings.**
- Engineered a complete **digital twin** for the "**Screw Picking Assembly Cell using AI**," a **factory automation task**. Performed **virtual commissioning** by designing mechanical components in **Fusion 360**, simulating the **mechatronic system** in **CoppeliaSim**, and validating control logic with **Python**.
- For the **Dooropen Robot** project, managed the complete **digital to physical workflow** by translating final **CAD designs** into manufacturing ready data (**sliced STL files**) and oversaw the **rapid prototyping** of over 150 functional components using Raise 3D printer, bridging the gap between **digital engineering** and physical implementation.

Skills

CAD-Proficiency
- NX SIEMENS
- FUSION 360
- Catia V5
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Fusion 360
- Matlab/Simulink
- ANSYS STRUCTURAL & FLEUNT
- CoppeliaSim
- Solidworks

Software-Proficiency
- PYTHON
- C/C++
- MS Office Suit

PLM Software - Proficiency
- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Academic Recognization, *RWTH*
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering -RWTH Aachen

vallabh.pataneni@rwth-aachen.de +49 17685944047 Bahnhofstraße 59/2, Renningen

linkedin.com/in/vallabh-pataneni

Summary

A Master's student in Automotive Engineering with a strong foundation in **Digital Engineering** principles and practical experience in **mechatronic system** development. My background includes leading a hybrid vehicle team for the Shell Eco-marathon and extensive **hands-on project work** involving **testing**, **prototyping**, and implementing digital solutions like **Product Lifecycle Management (PLM)**, **Digital Twin** creation, and **Virtual Commissioning**. I possess extensive experience with a wide range of industry-standard **CAD software** including **Catia V5**, **Siemens NX**, and **Fusion 360**. I am currently seeking a **mandatory internship** where I can apply my skills in digital factory automation and contribute to meaningful engineering challenges at Robert Bosch Manufacturing Solutions GmbH.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems), Mechatronics, DFM, FEM, Machine Learning, Addtive Manufacturing.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

Ecogenium e.V, *Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle.
- Perfomed the complete **CAD design**, **Kinematic & FEM simulation**, and **manufacturing** of both front (**double wishbone**) and rear (**trailing arm**) suspension systems for the "Katara" vehicle, specifically aiming for **minimal weight**, **optimal kinematic characteristics** by leveraging Generative Design using **Siemens NX**.
- Managed the complete **Product Lifecycle Management (PLM)** workflow for the “Katara” race car’s suspension and steering systems using **Siemens Teamcenter**, creating a **central hub** for all engineering data from initial **CAD** and **simulation** files to the final **Bill of Materials (BOM)** from manufacturing partners.
- Designed a compact **fuel cell test rig**, focusing on optimizing the packaging density and integration of the fuel cell and its associated electronic systems.
- Executed the CAD-to-manufacturing process for the **carbon fiber steering wheel**, overseeing its production using the **resin infusion method** and optimizing the composite layup for a **high stiffness-to-weight ratio.**

Experience

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- **Designed, simulated, prototyped,** and rigorously tested a **user-centric door-opening robot**, focusing on its **dynamics and mechanical reliability** using iterative design and simulation strategies in **Fusion 360**.
- Spearheaded the **Product Lifecycle Management (PLM)** for the **3rd Generation Dooropen Robot**, a complex **mechatronic system**, using **Autodesk Fusion 360 Manage** to create a **single source of truth** integrating multi disciplinary data including **3D CAD models**, electronic Data sheets and User manuals,**FEM/thermal** simulation results and **Manufacturing Drawings.**
- Engineered a complete **digital twin** for the "**Screw Picking Assembly Cell using AI**," a **factory automation task**. Performed **virtual commissioning** by designing mechanical components in **Fusion 360**, simulating the **mechatronic system** in **CoppeliaSim**, and validating control logic with **Python**.
- For the **Dooropen Robot** project, managed the complete **digital to physical workflow** by translating final **CAD designs** into manufacturing ready data (**sliced STL files**) and oversaw the **rapid prototyping** of over 150 functional components, bridging the gap between **digital engineering** and physical implementation

Prototyping Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- Contributed to the **conceptualization** and **prototyping** of a **proof-of-concept air taxi (scaled model)** using **SolidWorks**, with a focus on translating advanced mobility ideas into **functional designs**.
- Developed precise **CAD models** for complex **flight-critical components** such as wings and the landing mechanism, balancing **structural integrity** with lightweight design through **iterative simulation**.
- Performed **CFD simulations** on **fuselage** and **multiple aerofoil designs** to optimize geometry for improved lift and minimized drag.
- Developed distinct, custom **slicer interfaces using Python** for various 3D printing technologies, creating separate workflows for **High-Speed FDM, standard FDM, and SLA printers**.

Skills

CAD-Proficiency
- NX SIEMENS
- FUSION 360
- Catia V5
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Fusion 360
- Matlab/Simulink
- ANSYS STRUCTURAL & FLEUNT
- CoppeliaSim
- Solidworks

Software-Proficiency
- PYTHON
- C/C++
- MS Office Suit

PLM Software - Proficiency
- Autodesk Fusion 360 Manage
- Siemens Teamcentre

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Academic Recognization, *RWTH*
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linkedin.com/in/vallabh-pataneni

Summary

A **highly enthusiastic** Automotive Engineering Master's student seeking a mandatory internship in PEM fuel cell development. As the **leader** of a multidisciplinary team developing a **fuel cell powered vehicle** for the Shell Eco Marathon, I possess comprehensive, hands-on experience managing the **full lifecycle of performance optimization** for fuel cell systems. This includes conducting **experiments**, developing **optimization strategies** using **MATLAB/Simulink**, and leveraging extensive experience in **data analysis** and visualization with **Power BI and Python**. I am eager to apply these skills to advance fuel cell technology for heavy-duty vehicle applications at Robert Bosch.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems), Alternative Energy Systems,
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

Ecogenium e.V, *Team lead & member of pit crew 24/25* 06/2024 – Present | Aachen, Germany

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle.
- Conducted critical **experimental tests** on the **PEM fuel cell**, including **purging and short-circuiting**, while managing **telemetry data collection**, **visualization**, and in-depth **technical analysis** to drive **performance enhancements**.
- Leveraged **Python and Simulink** to create and deploy sophisticated, **data-driven driving strategies**, which was a critical factor in the team securing a **runner-up position** at the 2025 Shell Eco-Marathon.
- **Engineered and fabricated** a custom, compact **test rig** and its dedicated **thermal management system**, while using **Model-in-the-Loop (MIL) simulations** to **validate and refine** control strategies prior to implementation.
- Performed **Kinematic & FEM simulation** for the **vehicle's suspension systems**, demonstrating expertise in dynamic systems modeling.

Experience

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- **Designed, simulated, prototyped,** and rigorously tested a **user-centric door-opening robot**, focusing on its **dynamics and mechanical reliability** using iterative design and simulation strategies in **Fusion 360**.
- **Developed** and **implemented precise acceleration** and **deceleration curves** for **robot motion** by programming a **double H-bridge motor controller** using **Matlab/simulink**, controlling a **NEMA 23 stepper motor** integrated with a planetary gearbox.
- Successfully implemented **speech recognition AI** functionality, leveraging **Power BI** and **Python** to analyze **real-time data** and improve the efficiency of **command-based operations**.

Prototyping Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- Developed distinct, custom **slicer interfaces using Python** for various 3D printing technologies, creating separate workflows for **High-Speed FDM, standard FDM, and SLA printers**.
- Contributed to the **conceptualization and prototyping** of a proof-of-concept air taxi, fabricating critical components and optimizing designs for maneuverability and structural integrity.

Skills

Simulation-Proficiency
- Matlab/Simulink
- STAR CCM+
- ANSYS STRUCTURAL & FLEUNT
- Scilab
- Solidworks

Software-Proficiency
- PYTHON
- C/C++
- MS Office Suit

Python-Libraries
- NumPy
- Pandas
- PyTorch
- TensorFlow

CAD-Proficiency
- CATIA V5
- FUSION 360
- NX SIEMENS
- PTC CREO
- SOLIDWORKS

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

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linkedin.com/in/vallabh-pataneni

Summary

Ambitious and **solutions-oriented** Automotive engineering student seeking a **mandatory internship** to further advance my expertise in the development of **innovative mobility technologies**. I bring **hands-on experience** in programming, testing, and integrating a range of **motors** for specialized applications, leveraging **MATLAB/Simulink** for dynamic system modeling and **C/C++** for practical execution. As a **leader** of motorsport vehicle teams, I have successfully coordinated projects and motivated **multidisciplinary groups** to deliver high-performance results under demanding conditions. Driven by a **passion** for creating **unique mobility solutions**, I bridge **analytical rigor** with real-world testing and teamwork to achieve reliable, cutting-edge advances. Eager to contribute my **technical skills**, **leadership abilities**, and **commitment** to vehicle development at **Bosch eBike Systems**.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany
RWTH Aachen University
Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems), Mechatronics, Simulation Sciences, Mobile Propulsion Systems.
GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

Ecogenium e.V, 06/2024 – Present | Aachen, Germany
Mechanical Team lead & member of pit crew 24/25

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle.
- **Contributed to the selection of Permanent Magnet Synchronous Motors (PMSM)**, further developed **motor test benches to optimize efficiency**, and **integrated PMSMs into the vehicle** for optimal powertrain performance.
- **Performed Model-in-the-Loop (MIL) simulations in Simulink** to enhance the vehicle's control system performance and optimize operating strategies, directly applicable to **motor control simulation**.
- **Performed Kinematic & FEM simulation** for the vehicle's suspension systems, demonstrating expertise in **dynamic systems modeling**.
- Developed and iterated optimized driving strategies and control algorithms, leveraging **MATLAB/Simulink** and **Python**.

Experience

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- **Designed, simulated, prototyped, and rigorously tested a user-centric door-opening robot**, focusing on its **dynamics and mechanical reliability** using iterative design and simulation strategies in **Fusion 360**.
- **Developed and implemented precise acceleration and deceleration curves for robot motion** by programming a **double H-bridge motor controller using Arduino IDE**, controlling a **NEMA 23 stepper motor** integrated with a planetary gearbox.
- **Programmed the notes of "Pirates of the Caribbean" using Arduino IDE and a normal buzzer** for the robot, enabling it to play music upon activation, demonstrating practical application of **microcontroller programming** and **sound design**.
- **Developed and operated custom test benches to validate motor clampability** on various materials (Aluminum, carbon fiber, steel) to optimize weight, durability, and system integration, providing crucial data for iterative design refinement.

Prototyping Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- Contributed to the **conceptualization and prototyping of a proof-of-concept air taxi (scaled-down model)** using SolidWorks, gaining experience in translating innovative mobility solutions into tangible designs.
- **Executed CFD simulations** on the fuselage and aerofoil designs using **ANSYS Fluent** and **STAR CCM+** to optimize aerodynamic performance.
- Contributed to the **integration of E-motors** into the landing mechanism and as engine substitutes, providing foundational experience with **electric drive components**.
- Utilized an **industry standard dual-nozzle FDM rapid prototyping 3D printer (Raise 3D)** for **fabrication and continuous design refinement** of **over 150 different parts** based on testing feedback.

Skills

CAD-Proficiency
- CATIA V5
- FUSION 360
- NX SIEMENS
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- Matlab/Simulink
- STAR CCM+
- ANSYS STRUCTURAL & FLEUNT
- Scilab
- Solidworks

Software-Proficiency
- PYTHON
- C/C++
- MS Office Suit

Rendering and Visualization
- Keyshot
- Blender

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Paper Publication, *JNTUH*
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Bachelor's Thesis, *JNTUH-CEH*
CFD-Based Aerodynamic and **Spray System Design** for Enhanced Performance in **Agricultural Drones**
- Spearheaded aerofoil design with iterative **CFD analysis in ANSYS Fluent**, optimizing drone blade **lift** and **controllability**.
- Engineered and simulated a custom fertilizer spray nozzle using **Ansys Fluent**, performing internal/external flow simulations,methodology transferable to in-cylinder CFD for powertrain applications.

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linkedin.com/in/vallabh-pataneni

Summary

A highly motivated Automotive Engineering Master's student **seeking a mandatory internship**, with proven experience leading **motorsport vehicle projects** through the entire development lifecycle, from conceptual design and **Structural, Thermal, and CFD simulation** to hands-on fabrication and performance testing. My specialization in **vehicle thermodynamics and braking systems**, combined with over **300 hours of CFD analysis** using platforms like **STAR-CCM+ and Ansys Fluent**, positions me to excel. I am eager to apply my skills in **thermodynamic modeling and data analysis** to contribute to the **aerodynamic development of high-performance racing cars at BMW**.

Education

Master of Science in Automotive Engineering, 09/2023 – 06/2026 | Aachen, Germany

RWTH Aachen University

Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems), CFD, DFM, Thermodynamics, Additive Manufacturing

GPA: 1.7

Bachelors in technology Mechanical Engineering, 06/2019 – 04/2023 | Hyderabad, India

Jawaharlal Nehru Technological Univerisity

GPA: 1.3

Student Club

Ecogenium e.V, 06/2024 – Present | Aachen, Germany

Mechanical Team lead & member of pit crew 24/25

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the Shell Eco-Marathon, managing the **complete lifecycle** from concept to competition-ready vehicle.
- **Led the simulation-driven design process for "Katara's" aeroshell**, utilizing **Star-CCM+** to iteratively refine surface contours and component alignment, optimizing **aerodynamic performance** and achieving a **12% reduction in overall drag coefficient**.
- **Conducted wind tunnel testing on a scaled-down model of "Katara"** to validate aerodynamic predictions by comparing simulations with actual experimental data.
- **Analyzed wake dynamics, flow separation, and pressure distribution**, delivering actionable recommendations that enhanced airflow efficiency and led to measurable improvements in competition rollout energy efficiency.
- Performed the **complete design, Kinematic & FEM simulation** for both **front (double wishbone) and rear (trailing arm) suspension systems** for the "Katara" vehicle, specifically aiming for **minimal weight**, **optimal kinematic characteristics** for stability and maneuverability, and **efficient packaging** within the Eco-Marathon vehicle's constraints.
- Leveraged **Model-in-the-Loop (MIL) simulations** in **Simulink**, using competition data to optimize and validate vehicle **operating strategies**, which significantly enhanced **control system performance** and overall **vehicle efficiency**.

Experience

Prototyping Intern, *Garuda 3D* 01/2023 – 04/2023 | Hyderabad

- **Contributed to the conceptualization and prototyping of a proof-of-concept air taxi (scaled-down model) using SolidWorks**, focusing on translating innovative mobility solutions into tangible designs.
- **Executed CFD simulations on the fuselage using STAR CCM+** and other tools, systematically studying pressure distribution, drag, and lift while iterating with multiple aerofoil designs to refine performance, evaluating and optimizing trade-offs between stability, control, and efficiency.
- Performed the **design, simulation,** and **fabrication of** critical air taxi components **(e.g., wings, landing mechanism),** specifically **optimizing** for **maneuverability and structural integrity.**

Industrial Design Intern, *Techolution* 05/2023 – 08/2023 | Hyderabad

- **Designed and simulated various thermal sinks to reduce heat generated by motors**, demonstrating practical application in **thermal management system engineering** for electric drive subcomponents.
- **Designed, simulated, prototyped, and rigorously tested a user-centric door-opening robot**, focusing on its dynamics and mechanical reliability using iterative design and simulation strategies in Fusion 360.
- Performed numerous **Static and Buckling FEM analyses using Ansys** to design a weight-optimized chassis and motor housing.
- Utilized an **industry standard dual-nozzle FDM rapid prototyping 3D printer (Raise 3D)** for **fabrication and continuous design refinement** of **over 150 different parts** based on testing feedback.

Skills

CAD-Proficiency
- CATIA V5
- FUSION 360
- NX SIEMENS
- PTC CREO
- SOLIDWORKS

Simulation-Proficiency
- STAR CCM+
- ANSYS STRUCTURAL & FLEUNT
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Software-Proficiency
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Rendering and Visualization
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Languages

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- Engineered and simulated a custom fertilizer spray nozzle using **Ansys Fluent**, performing internal/external flow simulations,methodology transferable to in-cylinder CFD for powertrain applications.